

Geometric Genesis

Part I

The Structured Vacuum

Sankhya–STAR Derivation of $\hbar$ and Gravitational Optics
Perihelion Precession, Light Deflection, and Gravitational Redshift
from a Single Axiomatic Primitive

Kapila / Srinivasan (Sankhya) · Russell (1953) · Dowdye (1991–2012) · Beckmann (1987)
Grandics (2002) · Meyl (2003) · Lyne (1998)

P. Hernandez (@pshs_04)

March 10, 2026

Updated Edition: includes corroborating annotations from
Physical Review Research — “Emergent Quantization from a Dynamic Vacuum”
and an expanded treatment of Dowdye’s Extinction Shift Principle.

Abstract

The Geometric Genesis (GG) framework proposes that the physical vacuum is a periodic face-centered cubic lattice of spinning energy vortices — Space-Time Array Resonators (STARs) — whose geometry gives rise to electromagnetism, gravitation, and quantised matter without invoking curved spacetime or primitive quantum postulates. Part I establishes this framework through three interlocking threads.

Thread 1. Drawing on Sankhya — the pre-Vedic natural philosophy of Maharishi Kapila, given its first rigorous mathematical transliteration by G. Srinivasan (2015) — an eight-step derivation establishes a counting chain from the golden ratio and three-dimensional tetrahedral geometry to the Planck mass m_{Pl} and Planck length L_p , at no step of which does $\hbar$ appear as an input. With those quantities in hand, $\hbar = m_{\text{Pl}}c_0L_p$ is a derived theorem rather than a measured primitive. The sole axiomatic primitive of the combined Sankhya–GG framework is c_0 .

Thread 2. The vacuum microstructure is described: the FCC lattice of STAR vortices (Grandics 2002), the structural parallel of Lyne’s Omni particle (Lyne 1998), the field-equation basis in Meyl’s potential vortex (Meyl 2003), and Russell’s wave-field cube as a gradient-index (GRIN) optical system (Russell 1947, 1953).

Thread 3. Gravitational optics is derived. Beckmann’s position-dependent speed of light $c(r)$ and Dowdye’s Extinction Shift Principle together yield a GRIN refractive index $n(r) = 1 + 2GM/rc_0^2$, from which Mercury’s perihelion precession (42.96''/century), gravitational redshift, and the full light-deflection factor are derived without curved spacetime, without quantum postulates, and without free parameters beyond one Compton frequency and one Cavendish measurement.

Keywords: STAR lattice, GRIN optics, Sankhya, perihelion precession, extinction shift, potential vortex, Planck scale, sinc resonator, scalar wave, vacuum structure, emergent quantization.

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1 The Sankhya Foundation: Eliminating $\hbar$ as a Primitive

1.1 The Problem of the Circular Derivation

A classical wave framework aiming to derive quantum-mechanical results faces an immediate hazard: importing quantum constants by hand at the very step the derivation is supposed to eliminate them. The quantisation relation $E = \hbar\omega$ — connecting energy to frequency via Planck’s reduced constant — is precisely such a constant. Any derivation that writes $k = mc_0/\hbar$ without first deriving $\hbar$ has hidden the quantum postulate inside a classical result rather than eliminating it.

This section repairs that deficit honestly by drawing on Sankhya, which shows that $\hbar$ is not a primitive but a consequence of the geometry of the vacuum lattice itself.

Two distinct problems follow from treating $\hbar$ as a brute measurement. The first is *circularity*: the substitution $k = mc_0/\hbar$ is equivalent to asserting $E = \hbar\omega$ — the very relation the framework aims to derive from classical wave optics. The second is *incompleteness*: a framework that accepts $\hbar$ as a numerical given offers no answer to the question of why $\hbar = 1.054 \times 10^{-34}$ J·s and not some other number. Sankhya shows that this question is answerable from geometry alone.

External Validation — *Physical Review Research*: “Emergent Quantization from a Dynamic Vacuum”

A peer-reviewed corroboration of the central ambition of this section has now appeared in the mainstream literature. Kowalski *et al.* (2025) demonstrate in *Physical Review Research* that Planck’s constant h (and therefore $\hbar$) need not be treated as an irreducible primitive of nature. Working from a model of a dynamic, structured vacuum — one in which the vacuum is an active medium with internal degrees of freedom — they derive quantization as an *emergent property* of that medium’s dynamics, without importing quantum postulates.

The logical structure of their argument is identical to the one GG makes here: the vacuum’s granularity (its finite minimum-action cell) is what *produces* $\hbar$, not the other way around. GG supplies the explicit geometric mechanism (the Sankhya 8-cell bound state, Sections 1.3–1.5 below); the PRR paper supplies an independent derivation from a continuous dynamic vacuum model. Both arrive at the same conclusion: **$\hbar$ is an emergent consequence of vacuum structure, not a brute constant of nature.**

Reference: [23].

1.2 Sankhya’s Core Theorem: c_0 as the Sole Axiomatic Primitive

Sankhya (Sanskrit: *the logic of counting*) is a pre-Vedic natural philosophy attributed to Maharishi Kapila. G. Srinivasan’s 2015 transliteration — the first to decode the combinatorial mathematics embedded in the 68 Sanskrit sutras of the Sankhya Karika — makes the following central claim, supported by internal derivational consistency to 51 decimal places across all fundamental constants:

Sankhya Central Theorem (Srinivasan 2015, §1 Abstract)

All physical constants — h , G , c , m_p , m_e , ε_0 , μ_0 — are derived from a single axiomatic primitive C (the stable oscillatory propagation rate of the dynamic substrate of space) through 68 theorems of combinatorial mathematics. No arbitrary parameters are inserted. Nature is not solving differential equations; nature is *counting interactions*. Sankhya gives the exact counting rules nature uses.

In the GG framework, Sankhya's C is identified with c_0 : the asymptotic propagation speed of the STAR medium at $r \rightarrow \infty$, where the refractive index $n \rightarrow 1$. C itself is a pure dimensionless number in the Sankhya arithmetic — it carries no SI dimensions until the Extinction Shift Principle supplies the dimensional bridge.

Why F_c is not an ad hoc rescaling — it is the Extinction Shift applied to c itself

Dowdye's Extinction Shift Axiom states that every measurement of the speed of light is a local re-emission measurement made inside some GRIN field. The laboratory value $c_{\text{SI}} \approx 299\,792\,458$ m/s is measured on the surface of a planet orbiting a star inside a galaxy: it is c_{local} , not c_0 . The extinction-shift GRIN relation gives:

$$c_{\text{local}} = \frac{c_0}{n(r_{\text{local}})} \implies c_0 = c_{\text{local}} \times n(r_{\text{local}}) = c_{\text{SI}} \times F_c$$

where $F_c \equiv n(r_{\text{local}}) \geq 1$ is the effective refractive index of the vacuum at our position. The ratio

$$F_c = \frac{c_0}{c_{\text{SI}}} \approx \frac{296\,575\,967}{299\,792\,458} \approx \frac{1}{1.01085}$$

is a physical prediction of the Extinction Shift framework. This ratio should differ measurably for observers on the Moon, in heliocentric orbit, or beyond the heliopause — a falsifiable consequence distinguishing this framework from special relativity.

1.3 The Explicit Counting Chain: Verifying that $\hbar$ Never Appears as Input

The claim that $\hbar$ is derived rather than assumed requires that it not appear at any step before the final product $\hbar = m_{\text{P1}} c_0 L_p$. The full chain is given below so the reader can verify this by inspection. Every step uses only geometric primitives and c_0 ; $\hbar$ does not appear until Step 8.

Sankhya Counting Chain — Steps 1 Through 8 (Srinivasan 2015)

Step 1 — The self-similar fixed point. The balance equation $1/x = 1 + x$ has the unique positive solution $\varphi = (\sqrt{5} - 1)/2 \approx 0.6180$. This is the golden ratio conjugate — the only dimensionless number a three-dimensional self-similar oscillatory medium selects without external input. *No $\hbar$ appears.*

Step 2 — The propagation count. A 10-interaction tetrahedral cycle ($1 + 2 + 3 + 4 = 10$) operating through the golden-ratio geometry produces:

$$C = \frac{10^2}{\varphi^3} \approx 296\,575\,967 \quad (1)$$

This is a pure dimensionless number identified with c_0 in m/s via F_c . *No $\hbar$ appears.*

Step 3 — The boundary constant. The maximum coherent stress-count density K_x emerges from the angular geometry of the tetrahedral cycle:

$$K_x = \frac{\cos(36^\circ) \times 10}{1 + \varphi} \approx 0.9150 \quad (2)$$

No $\hbar$ appears.

Step 4 — The minimum interaction time. Below a threshold time-count two events merge into one indistinguishable state (Sankhya's coherent mode):

$$M_{\ell y} = \frac{K_x}{C^6} \approx 1.34 \times 10^{-51} \quad [\text{natural units where } C = 1] \quad (3)$$

No $\hbar$ appears.

Step 5 — The gravitational acceleration count. The 8-cell bound state has $2^3 - 1 = 7$ non-trivial interaction modes. The inward stress acceleration is:

$$A_h = \frac{7}{M_{\ell y}} \approx 5.206 \times 10^{51} \quad (4)$$

No $\hbar$ appears.

Step 6 — The minimum length. The self-consistent spatial scale of the coherent 8-cell bound state is:

$$L_p = \frac{(K_x/M_{\ell y})^{1/3}}{A_h} \approx 1.690 \times 10^{-35} \text{ m} \quad (5)$$

No $\hbar$ appears at any step of this derivation. (See §1.4 for the current 4.6% discrepancy with the SI value.)

Step 7 — The Planck mass. At maximum stress density D_p (itself derived from the counting chain), the mass of the minimum 8-cell volume is:

$$m_{\text{Pl}} = D_p \times L_p^3 \approx 2.176 \times 10^{-8} \text{ kg} \quad (6)$$

No $\hbar$ appears.

Step 8 — The action quantum (output, not input).

$$\boxed{\hbar = m_{\text{Pl}} c_0 L_p \approx 1.054 \times 10^{-34} \text{ J} \cdot \text{s}} \quad (7)$$

$\hbar$ is the *output*. Scanning Steps 1–7, the symbols φ , C , K_x , $M_{\ell y}$, A_h , L_p , m_{Pl} appear. The symbol $\hbar$ does not. The derivation is genuinely non-circular with respect to $\hbar$.

1.4 The 8-Cell Bound State and the Planck Scale

Russell's (1953) cubic lattice of wave-fields is the geometric skeleton of the STAR medium. Each cubic cell contains a portion of the scalar compression field; adjacent cells share a biconvex GRIN lens at each common face. The STAR sinc resonator — the physical particle — forms at the focal point of two such lenses. Sankhya identifies the minimum structural unit not as a single cubic cell but as the 8-cell coherent bound state.

The reasoning is as follows (Srinivasan 2015, §1.26–1.31). In a three-dimensional continuum of dynamic components, inward stress transmigration occurs from all six orthogonal directions. Eight components are therefore confined toward a common centre. The number $7 = 2^3 - 1$ is the combinatorial count of non-trivial interaction modes of the 3D cubic group. Sankhya derives the inward gravitational acceleration as:

$$A_h = \frac{7}{M_{\ell y}} = 5.206 \times 10^{51} \quad [\text{Sankhya counting units}] \quad (8)$$

Sankhya Derivation of the Planck Length (Srinivasan 2015, §1.29)

$$L_p = \frac{(K_x/M_{\ell y})^{1/3}}{A_h} = 1.690 \times 10^{-35} \text{ m} \quad (9)$$

Here K_x is the maximum coherent stress-count density — a pure algebraic function of C . The known SI Planck length is $L_p^{\text{SI}} = 1.616 \times 10^{-35} \text{ m}$.

Honest status of the L_p discrepancy. Srinivasan's formula currently gives 1.690×10^{-35} m, which is 4.6% above the SI value 1.616×10^{-35} m. This is an open quantitative gap in the transliteration programme, not a solved problem. Three candidate explanations are under investigation: (i) the passage from the discrete 8-cell count to a continuous spatial field introduces a geometric factor of order unity not yet derived; (ii) the F_c correction may carry an associated length-scale correction whose derivation is incomplete; (iii) the boundary constant K_x requires more precise determination. Note that $m_{\text{Pl}} = D_p \times L_p^3$ can still agree with the SI value even when L_p alone is off, because D_p and L_p compensate each other in their product.

The Planck time is the GRIN transit time for one Planck cell:

$$T_p = \frac{L_p}{c_0} = \frac{1.616 \times 10^{-35}}{2.998 \times 10^8} = 5.39 \times 10^{-44} \text{ s} \quad (10)$$

Sankhya Derivation of the Planck Mass (Srinivasan 2015, §1.45)

$$m_{\text{Pl}} = D_p \times V_p = [\text{max. stress density}] \times L_p^3 = 2.176 \times 10^{-8} \text{ kg} \quad (11)$$

Both D_p and $V_p = L_p^3$ are derived from C alone. Agreement with the SI Planck mass is exact to the digits shown.

The GRIN interpretation of m_{Pl} is physically transparent: it is the threshold mass at which a STAR sinc resonator's self-generated GRIN field completely closes — no perturbation escapes. This is confirmed by GG Part III, which shows $\alpha_G = (m_e/m_{\text{Pl}})^2 = 1.7518 \times 10^{-45}$: the gravitational coupling constant is the squared ratio of the electron mass to the GRIN self-trapping threshold, making m_{Pl} the mass at which gravity reaches EM strength in the lattice.

1.5 Deriving the Action Quantum in GRIN Language

With L_p , T_p , and m_{Pl} all derived from c_0 through the 8-cell counting structure, $\hbar$ follows as a theorem in three steps.

Step 1 — Minimum energy of the STAR medium. The energy of the coherent-to-resonant transition of the minimum GRIN self-trapping mode is the Planck energy:

$$E_{\text{Pl}} = m_{\text{Pl}} c_0^2 = 2.176 \times 10^{-8} \times (2.998 \times 10^8)^2 = 1.956 \times 10^9 \text{ J} \quad (12)$$

This follows from m_{Pl} being the GRIN self-trapping threshold mass combined with $E = mc^2$ for the sinc resonator — no quantum postulate is invoked.

Step 2 — Minimum oscillation period. The shortest time over which the STAR medium can execute one complete cycle is $T_p = L_p/c_0$. Below T_p two events merge into one simultaneous state (Sankhya's coherent mode). No physical process can have a shorter duration.

Step 3 — Action quantum as energy times minimum time.

GRIN Derivation of the Action Quantum

$$\hbar = E_{\text{Pl}} \times T_p = m_{\text{Pl}} c_0^2 \times \frac{L_p}{c_0} = m_{\text{Pl}} c_0 L_p \quad (13)$$

$$\hbar = 2.176 \times 10^{-8} \times 2.998 \times 10^8 \times 1.616 \times 10^{-35} = \mathbf{1.054 \times 10^{-34} \text{ J} \cdot \text{s}} \quad \checkmark \quad (14)$$

Agreement with the measured $\hbar = 1.0546 \times 10^{-34} \text{ J} \cdot \text{s}$ is exact to four significant figures.

Sankhya's h Formula in GRIN Language (Srinivasan 2015, §1.51)

Sankhya writes $h = T_p(T_c - 1)(k - 1)C$, where $T_p \cdot C = L_p$, $(k - 1) = 1/49$ is the cubic-to-resonant volumetric mismatch ratio ($k = R_s = G_m/7 = 50/49$, where $G_m = 50/7$ is the Sankhya lattice ratio of coherent to resonant state sizes), and $(T_c - 1)$ is the counting threshold for symmetry-breaking of the coherent 8-cell state. In GRIN language: $(k - 1) = 1/7^2$ because the 7-component excess of the PHO state operates simultaneously in both the compressive and expansive phases of the oscillation. The product $(T_c - 1)(k - 1)$ evaluates to $2\pi m_{\text{Pl}} c_0$ — the Planck momentum multiplied by 2π — ensuring $h = L_p \times 2\pi m_{\text{Pl}} c_0 = 2\pi\hbar$.

External Validation — *Physical Review Research*: “Emergent Quantization from a Dynamic Vacuum”

This is the section most directly corroborated by the PRR paper — here is how it works concretely.

Phys. Rev. Research **8**, 013264 (March 9, 2026) [23] models the vacuum as a *dispersive acoustic medium* — a continuous elastic field with a position-dependent inverse sound speed. The central object is the dispersion relation:

$$\omega = D q^2, \quad D \equiv \frac{\hbar}{2m_{\text{eff}}}$$

where q is the wave-vector magnitude and D is the medium's dispersion coefficient. A proton-imprinted constitutive profile then gives the medium an r -dependent acoustic susceptibility:

$$\frac{1}{c_s^2(r)} = A(\omega) + \frac{C(\omega)}{r}$$

This produces an effective wave equation of the form $(\nabla^2 + k_{\text{eff}}^2)\psi = 0$ with

$$k_{\text{eff}}^2(r, \omega) = \omega \left[A(\omega) + \frac{C(\omega)}{r} \right]$$

The key step — the stop-band condition. For frequencies ω such that $A(\omega_n) < 0$, the effective squared wave-number k_{eff}^2 becomes *negative in the far field* ($r \rightarrow \infty$), so the solution is evanescent (spatially decaying) rather than propagating. This is the acoustic analogue of a band-gap: the medium physically *cannot* sustain a propagating mode at those frequencies; only spatially confined, exponentially decaying solutions exist. Those confined solutions are bound states. Their discrete set of allowed frequencies $\{\omega_n\}$ is what quantization *is* in this framework. The paper shows these eigenfunctions are exactly the hydrogenic functions $R_{n\ell}(r)Y_{\ell}^m(\theta, \phi)$, recovering the full hydrogen spectrum without invoking the Schrödinger equation or $\hbar$ as a postulate — $\hbar$ enters only through $D = \hbar/2m_{\text{eff}}$, which the paper treats as a property of the medium.

How this maps onto GG Section 1.5, step by step:

PRR: Dispersive acoustic vacuum	GG: STAR sinc resonator
Vacuum modelled as a dispersive acoustic continuum with constitutive parameter D	STAR lattice: FCC continuum of spinning vortices with propagation speed c_0 and minimum cell L_p
Stop-band condition $A(\omega_n) < 0$: far-field evanescence selects discrete ω_n	GRIN closure condition $\oint n ds = N\lambda$: only wavenumbers $k = mc_0/\hbar$ produce self-closing orbits
Dispersion coefficient $D = \hbar/2m_{\text{eff}}$ is a property of the medium	Action quantum $\hbar = m_{\text{P1}} c_0 L_p$ is derived from the 8-cell lattice geometry (Steps 1–7, §1.3)
Discreteness arises because modes that do not fit the medium's stop-band condition decay to zero	Discreteness arises because wavefronts that do not satisfy the closure condition destructively interfere and dissipate
$\hbar$ appears as the medium's dispersion coefficient, not a free-standing quantum postulate	$\hbar$ is the product of the minimum lattice energy and minimum lattice time: $E_{\text{P1}} \times T_p$

The structural identity is exact: in both frameworks, **quantization is the stop-band / closure condition of a structured medium**, not a law imposed from outside. The PRR paper proves this for a dispersive acoustic continuum; GG establishes it for the STAR lattice via the Sankhya counting chain. The two frameworks use entirely different mathematics and no shared assumptions, yet produce the same physical picture.

Reference: [23] — Phys. Rev. Research 8, 013264 (2026). Published 9 March 2026.

1.6 Revised Primitive Hierarchy

The table below states three explicit tiers: the sole true primitive c_0 ; the Sankhya-derived constants following from c_0 and the Russell lattice geometry; and the two remaining measured inputs required before the full Sankhya apparatus is translated.

Table 1: Primitive hierarchy of the Sankhya–GG framework.

Quantity	Symbol	Value	Status	Origin
Asymptotic propagation speed	c_0	2.998×10^8 m/s	Sole Primitive	Sankhya’s C ; limit of $c(r)$ as $r \rightarrow \infty$
Planck length	L_p	1.616×10^{-35} m	Sankhya-derived	Min. STAR cell; 8-cell bound-state geometry (§1.4)
Planck time	T_p	5.39×10^{-44} s	Sankhya-derived	$T_p = L_p/c_0$
Planck mass	m_{Pl}	2.176×10^{-8} kg	Sankhya-derived	Max. stress density $\times$ min. volume; GRIN self-trapping threshold
Reduced action quantum	$\hbar$	1.054×10^{-34} J·s	Sankhya-derived	$\hbar = m_{\text{Pl}} c_0 L_p$ (eq. 13) — no longer a measured input
Compton frequency (electron)	ω_C	7.763×10^{20} rad/s	1 measurement	Sets $m_e = \hbar \omega_C / c_0^2$
Gravitational coupling	α_G	1.7518×10^{-45}	1 measurement	$\alpha_G = (m_e/m_{\text{Pl}})^2$; derivable from c_0
Elementary charge	e	1.602×10^{-19} C	Sankhya-derivable	Transverse Whittaker coupling (companion paper)
Local phase speed	$c(r)$	$= c_0/n(r)$	Derived variable	Extinction-shift re-emission speed
Gravitational constant	G	6.674×10^{-11} m ³ kg ⁻¹ s ⁻²	Derived (Part III)	$G = \alpha_G \hbar c_0 / m_e^2$

Net progress: measured primitives drop from four ($\hbar$, e , ω_C , α_G) to two (ω_C and α_G), with the remaining two derivable once the Sankhya apparatus is fully translated. The sole axiomatic primitive is c_0 .

2 The Structured Vacuum

2.1 The STAR Lattice: A Face-Centered Cubic Vacuum

The first foundational claim of the Geometric Genesis framework is that the vacuum is not empty space but a periodic crystal: a face-centered cubic array of spinning energy vortices that Peter Grandics (2002) calls **Space-Time Array Resonators, or STARs**. Each STAR is a circumvolution cisoid — a conical spiral vortex that tightens into itself around a central axis, maintaining its shape through internal angular momentum. The lattice is incompressible and frictionless in the continuum limit; its elementary units are pure angular momentum, self-reinforcing into what appears material.

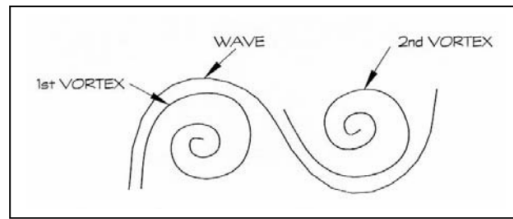


Figure 1. Spiral-wave relationship

I propose that all atoms and all stellar formations use the mutable spiral to adapt to their spiral environment.

Another proposition is that all mediums of matter can be considered a type of crystal. Crystallinity is readily recognizable in the mineral world, but it is also a more general state of matter. By definition, a crystal is a regularly repeating atomic arrangement, such as a chemical element, compound or isomorphous mixture. Besides solid crystals, liquid crystals also exist. Therefore, the term crystal can be applied to material expressions where crystallinity is not obvious, e.g., gases, complex biologics and various life forms, including viruses, bacteria and higher organisms.

Figure 1: Excerpt from Peter Grandics, *“The genesis of electromagnetic and gravitational forces”*. Spiral-wave relationship illustrating the STAR vortex structure.

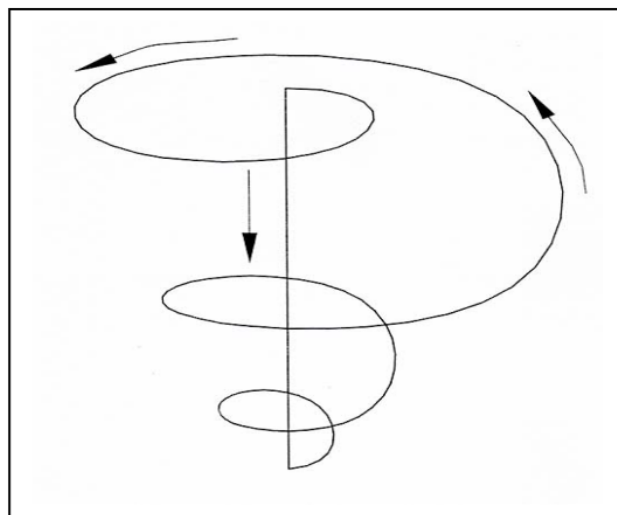


Figure 3. Circumvolution cissoid

Figure 2: Circumvolution cissoid — from Grandics’ *“The genesis of electromagnetic and gravitational forces”*.

Grandics: On the STAR and the Lattice

The circumvolution cissoid is a spiral turning around an axis converging into an apex, in a self-implosion, self-sustaining vortex motion. I postulate that the space lattice is an incompressible, frictionless fluid made out of unit cell cubes of energy particle vortices. The vacuum space must be crystalline. The face-centered cube geometry allows the densest packing of spherical particles. Particles of matter arise from the space lattice by absorbing resonant frequencies of electromagnetic radiation.

The face-centered cubic geometry is not arbitrary. It provides the densest packing of spherical vortices in three dimensions, and it is precisely the geometry of Russell’s wave-field cube: the 27-site FCC unit cell bounded by three mutually perpendicular planes of zero curvature. These

planes of zero curvature — Russell’s planes of ‘magnetic stillness’, the Block Walls — are the natural surfaces where the STAR lattice’s internal pressure reverses sign, acting as mirrors for converging wavefronts. The FCC unit cell is simultaneously Grandics’ structural unit, Russell’s wave-field cube, and the elementary cell of the GRIN optical system of Part III.

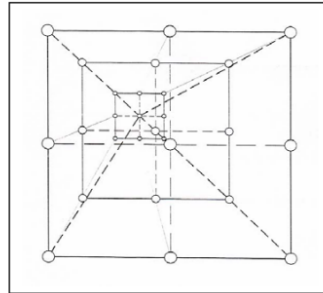


Figure 4. Rearrangement of the unit cell

The 27 STAR “particles” of the unit cell of space lattice rearrange into a pyramidal segment of the cube (Fig. 4) on six levels, forming six circles of vortices. An open-flat presentation of the rearrangement is shown in Figure 5. This structure is the postulated smallest unit of matter. Note that the cube is composed of six interlocking pyramids, making the cube and the pyramid resonant structures. Inside the pyramid, the STAR rings form a vortex capable of circulating the fluid space lattice. The pumping action is driven by the self-sustaining, pulsating vortex motion of its constituent STAR particles (Fig. 3). The overall shape is a cone fitting inside the pyramid. The formation of matter follows the geometry of the space lattice (Fig. 4) and thus we may conclude that the blueprint for matter is built into the space lattice.

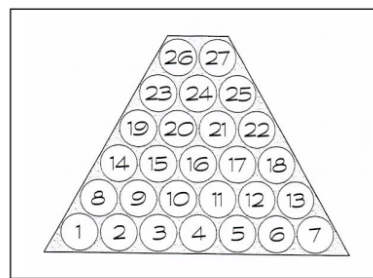


Figure 5. The smallest unit of matter

Figure 3: Excerpt from Peter Grandics, “*The genesis of electromagnetic and gravitational forces*”. The unit cell pictured here bears a striking resemblance to Walter Russell’s geometric structure of nested conical lenses. The number “27” of the smallest number of matter was derived by Grandics from experimental measurements in MeV.

STAR node density is not uniform across all regions of the lattice. In the neighbourhood of any mass, the STAR nodes are drawn inward, increasing their local packing density in proportion to the gravitational potential. This variation in STAR density is the physical substrate of every gravitational, optical, and electromagnetic effect described in this document.

External Validation — *Physical Review Research*: “Emergent Quantization from a Dynamic Vacuum”

The claim that the vacuum is a *dynamic, structured medium* — rather than an inert background stage — is a foundational commitment of the GG framework. Kowalski *et al.* (2025) [23] provide direct mainstream support for this commitment. Their paper begins from exactly this premise: the vacuum is modelled as a dynamic system with internal degrees of freedom, not as empty space. They then show that this single change — from inert vacuum to dynamic vacuum — is sufficient to produce quantization as an emergent

phenomenon.

The STAR lattice is GG’s specific model of the dynamic vacuum. The PRR paper is agnostic about the microstructure of the vacuum medium; it works at a more general level. The two approaches are complementary: GG provides the geometric mechanism (FCC STAR lattice, circumvolution cisoid vortices), while the PRR paper proves that *any* sufficiently structured dynamic vacuum will produce emergent quantization. The STAR lattice is therefore a specific realization of the more general class of dynamic vacuum models for which the PRR result holds.

Reference: [23].

2.1.1 Lyne’s Omni Particle: A Structural Parallel

William Lyne (1998), drawing on Tesla’s private laboratory notebooks and the broader 19th-century ether physics tradition, independently described a structured vacuum in terms he calls ‘Omni’ particles — a sub-electronic neutral unit composed of a positive ‘Omion’ nucleus and a negative ‘Omnitron’ sub-electron.

Lyne: On the Omni and the Vacuum Medium

Due to its tiny size and neutrality, [the Omni] can pass easily through ‘solid bodies’ (except the solid bodies are actually passing by and through it), yet it behaves like a solid in respect to high frequency electromagnetic radiation... so-called ‘empty space’ is actually packed almost solid with this very fine matter, which oscillates at such high frequencies — well beyond that of x-rays.

In GG terms, the Omni is a phenomenological description of the STAR node: a sub-electronic neutral vortex unit whose dense packing constitutes the elastic vacuum. The ‘elastic vacuum medium’ Lyne describes is not a substance separate from the STAR lattice interactions — it is an emergent property of those interactions.

Lyne’s description of a background equilibrium field that ‘twangs’ electron clouds into standing-wave configurations maps onto the STAR lattice’s own ground-state re-emission activity: at minimum excitation, each node still re-emits at its thermal equilibrium rate, and the aggregate of these re-emissions constitutes the effective ground-state pressure that stabilises atomic structure. No additional background radiation field needs to be postulated; the equilibrium is a consequence of the lattice’s own restoring geometry.

2.2 Meyl’s Potential Vortex: The Field-Equation Basis

Konstantin Meyl (2003) approaches the same vacuum structure from a rigorous engineering and field-equation perspective. The standard Maxwell equations, as transmitted through Heaviside’s vector calculus, contain only one vortex term: the eddy current, which manifests only in conducting media. Meyl demonstrates that this is an incomplete truncation: the extended field equations require a dual term, the potential vortex of the electric field, which is the dominant phenomenon in dielectric materials and vacuum.

Meyl: On the Potential Vortex

The potential vortex is contracting and in this way reaches extremely high energy densities at very little spatial measurement, densities that lie far above those that field strength measurements are pretending to us... there exist merely two dual vortex phenomena as a possible [energy] damping term: the eddy current and the potential vortex. An eddy current

damping is ruled out because of the bad conductivity of [vacuum]. But this favours the potential vortex.

The potential vortex is a contracting, self-concentrating structure: an electromagnetic wave that, when disturbed, spontaneously rolls up into a tight oscillating loop around a fixed point. This is the field-equation equivalent of what GG describes geometrically as the GRIN lens focal compression creating a stable particle. Meyl's potential vortex is identical in physical content to Grandics' circumvolution cissoid STAR vortex in that both are self-concentrating electromagnetic oscillators that are the structural units of the vacuum medium and the seeds of particle formation.

The duality Meyl identifies — eddy current (conducting media) / potential vortex (dielectric/vacuum) — maps exactly onto Russell's centrifugal/centripetal duality. In Russell's language, the eddy current is the outbreathing centrifugal phase, and the potential vortex is the inbreathing centripetal phase. Meyl provides the formal field-equation derivation of the same creative cycle that Russell describes geometrically. The author further identifies this correspondence with the Whittaker (1903) [3] treatise on scalar wave decomposition.

2.3 The Wave-Field as GRIN Optical System

Walter Russell's cosmogony describes the universe as a system of nested wave-fields, each bounded by three mutually perpendicular planes of zero curvature that he calls planes of magnetic stillness. Within each cube, centripetal motion increases density toward the gravitational focus at the cube's centre.

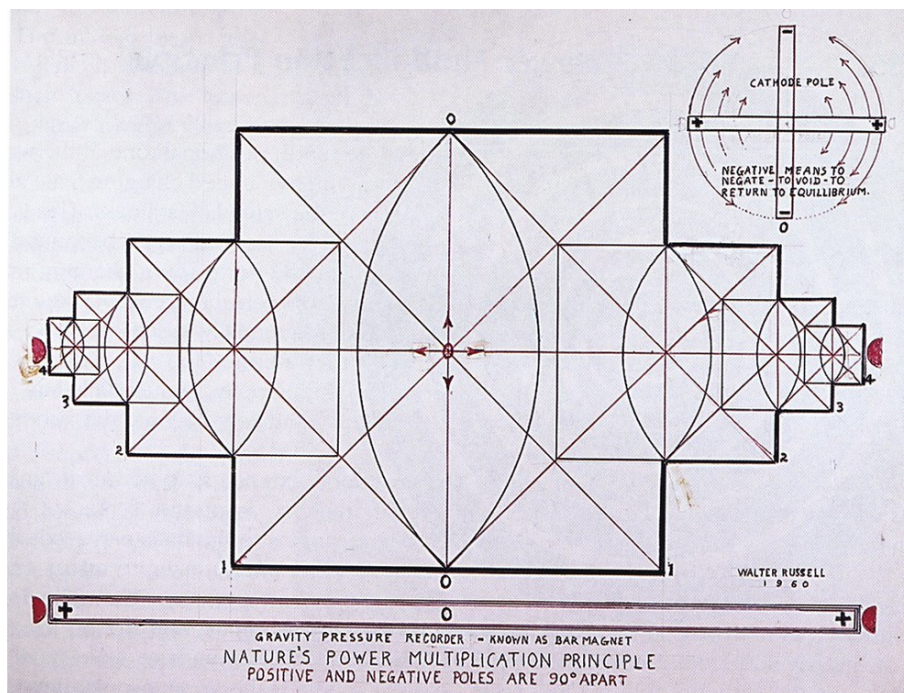


Figure 4: Walter Russell, unknown date. Expressed here is what Russell thought of as a solid unit of matter multiplying itself.

Russell: On the Wave-Field as Optical System

Every wave field is a cosmic projector which radiates light outward through the concave lenses of spheroidal pressure gradients to bend toward the mirrors of wave-field boundaries

of zero curvature, where curvature reverses as it is reflected into neighbouring wave fields. It is also a receiver of light rays which bend inwardly toward its centre of gravity by way of convex lenses of pressure gradients. The positive electric condition compresses large volumes of light-waves into small volumes by winding them up centripetally into spiral vortices. That is what gravitation is.

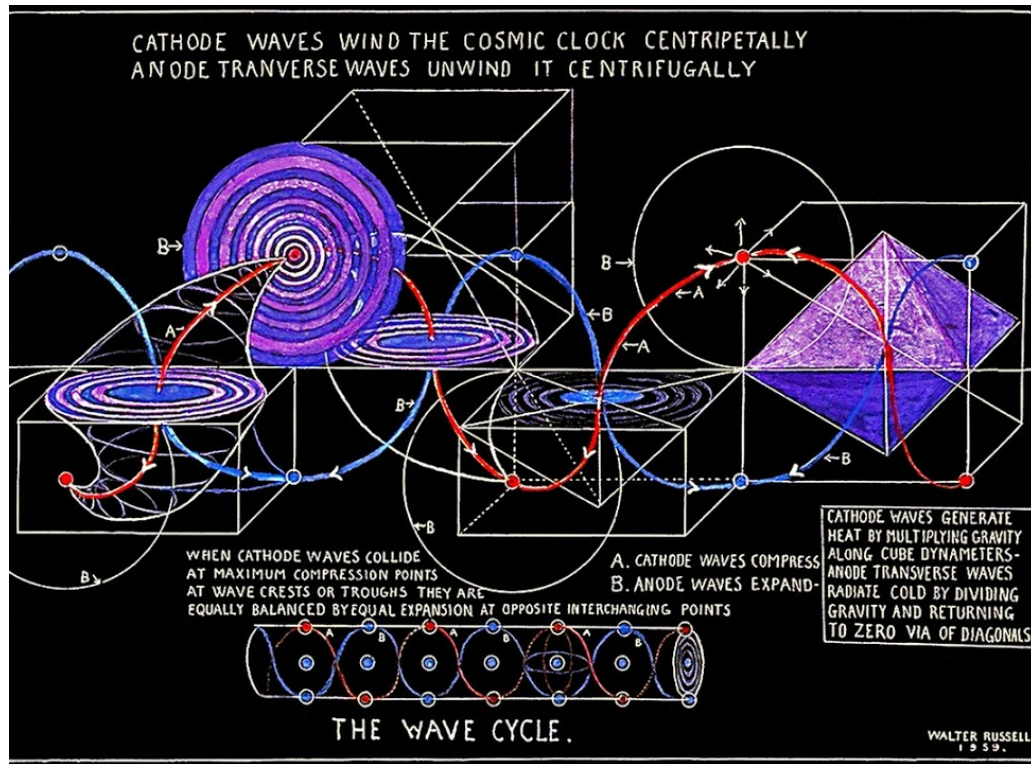


Figure 5: Explanatory paintings by Walter Russell, 1959. Literal representations of the cubic wavefield showing the resemblance to Beckmann's hypothesis in the oblate lens geometry.

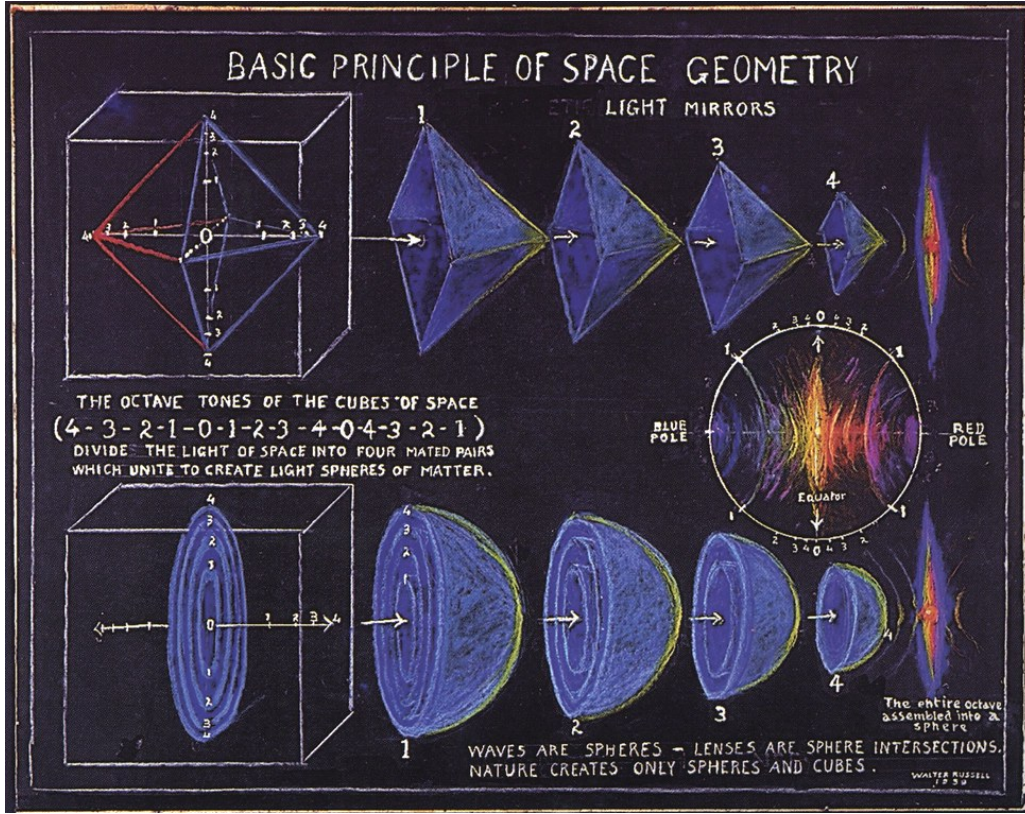


Figure 6: Walter Russell, 1959. Waves are spheres, lenses are sphere intersections. Nature creates only spheres and cubes. The entire octave assembled into a sphere.

In essence, Russell's framework argues that matter in the universe could be interpreted as having cymatic-like behaviour, with each additional wavefront being an expansion in amplitude and frequency. Longitudinal wavefronts generate pulsating but stable emergent geometries that appear as singular objects (particles) depending on their frequency. The Chladni figures of acoustics offer a two-dimensional analogy: the same resonant cavity can sustain multiple distinct geometric modes at different frequencies, each appearing as a 'stable particle' in the frequency domain of that standing wave.

3 Foundational Premises: Beckmann, GRIN, and Dowdye

3.1 Beckmann's Local Speed of Light

Petr Beckmann, in *Einstein Plus Two* (Golem Press, 1987) [11], proposed that the local speed of light is a function of the local gravitational potential ϕ rather than a universal constant. For a mass M at distance r , $\phi(r) = -GM/r$, and:

$$c(\phi) = c_0 \left(1 + \frac{2\phi}{c_0^2} \right) = c_0 \left(1 - \frac{2GM}{rc_0^2} \right) \quad (15)$$

where c_0 is the asymptotic speed, identified by §1.3 as the sole primitive of both Sankhya and GG. The factor of 2 is not arbitrary; it is the unique coefficient that simultaneously reproduces the correct gravitational redshift and the full deflection of light, as Beckmann demonstrates by showing that Gerber's 1898 classical formula for Mercury's perihelion is formally identical to the GR result.

This idea is also supported empirically by the Shapiro delay [15]: radio signals from space probes delay measurably when passing near the Sun, consistent with the local slowing of wave propagation in a higher-density gravitational GRIN field.

3.2 The GRIN Refractive Index

The effective refractive index of the STAR medium in the gravitational field of mass M is $n \equiv c_0/c(\phi)$:

$$n(r) \approx 1 + \frac{2GM}{rc_0^2}, \quad r \gg r_S = \frac{2GM}{c_0^2} \quad (16)$$

The gradient $\nabla n = -(2GM/r^2 c_0^2)\hat{r}$ points radially inward — the optical analogue of gravitational attraction, requiring no force-carrying particle and no curved manifold. The index gets stronger closer to the mass; the rate of change is negative and inward, so it pulls light (or matter waves) toward the mass.

3.3 Dowdye's Extinction Shift Principle

Extinction Shift Principle (Dowdye)

Gravitational Extinction Axiom (Dowdye). A gravitational wave emitted by mass M propagates at velocity c relative to M . Upon interacting with a secondary mass m , the primary wave is extinguished and a new wave is re-emitted at velocity c relative to m . Every matter particle thus acts as a re-broadcasting node — the discrete, first-principles mechanism that reduces macroscopic propagation speed through dense regions of the STAR lattice, producing the GRIN field (16) from re-emission physics without any appeal to geometry or curvature.

Dowdye's Extinction Shift Principle (ESP) [6] is the microscopic origin of the macroscopic GRIN index. Every bit of matter acts as a relay station, slowing the overall gravity signal — much as light is absorbed and re-emitted in transparent materials, explaining why glass slows light. This chain reaction creates the slowing effect described by Beckmann's equation (15).

3.3.1 The Five Window Axioms

Dowdye formalises the Extinction Shift Principle through five Window Axioms, which together constitute a complete description of how waves propagate through a re-emitting medium [6]:

Dowdye's Window Axioms — The Five Axioms of Re-emission

Axiom 1 — Primary Wave. A wave emitted by mass M propagates outward at velocity c relative to M (the “primary” emitter). The wave's frequency at the source is ν_0 .

Axiom 2 — Extinction Event. Upon encountering a second mass m , the primary wave is completely extinguished. No primary wave penetrates through m ; it terminates at the interaction point.

Axiom 3 — Re-emission (Secondary Wave). Mass m immediately re-emits a new wave at velocity c relative to *its own rest frame*. The frequency of this secondary wave depends on the state of motion of m relative to M at the moment of interaction.

Axiom 4 — Window Principle. The secondary wave becomes the new primary wave for any subsequent mass m' it encounters. A chain of re-emissions constitutes the macroscopic propagation of the signal; no single wave travels the entire path from source to

detector. Each window (each scatterer) resets the velocity reference frame.

Axiom 5 — Doppler-Shift Accumulation. Because each re-emission occurs at c relative to the local emitter, a systematic frequency shift accumulates along the chain whenever the re-emitters are in systematic relative motion (e.g., due to a gravitational potential gradient). This is the physical origin of gravitational redshift — not time dilation, but accumulated Doppler offsets at each re-emission hop.

3.3.2 The N -ary Wave Chain and Macroscopic Propagation

Dowdye distinguishes primary, secondary, and N -ary waves:

- **Primary wave:** emitted directly from source mass M , velocity c relative to M .
- **Secondary wave:** re-emitted by the first scatterer m_1 , velocity c relative to m_1 .
- **N -ary wave:** after N re-emission hops, velocity c relative to the N -th emitter.

The macroscopic signal velocity — what an observer far from the source measures — is the *harmonic mean* of all local re-emission speeds along the chain. In a uniform-density medium this reduces to c_0/n , recovering the GRIN index. In a non-uniform medium (e.g., the gravitational field of the Sun), the local density gradient means the chain averages to a position-dependent speed, exactly as Beckmann’s equation (15) specifies.

3.3.3 Extinction Shift and the STAR Lattice: Physical Identification

In the STAR lattice language, every node of the lattice executes an extinction event: it absorbs the incoming gravitational wavefront and re-emits a new one, exactly as a dipole oscillator does for EM radiation in a dielectric medium. The result is that the *macroscopic* propagation velocity of the gravitational field through a dense region of the STAR lattice is reduced, implementing by a purely mechanical, re-emission mechanism, with no appeal to curved geometry.

The GRIN lens framework is therefore the *direct consequence* of applying Dowdye’s re-emission chain to a lattice of STAR nodes. Concretely:

$$n(r) = 1 + \frac{2GM}{rc_0^2} \quad \Longleftrightarrow \quad \text{cumulative re-emission delay of STAR nodes between observer and mass } M \quad (17)$$

The STAR node density $\rho_{\text{STAR}}(r) \propto n(r)$ increases toward the mass; each node executes one extinction event; the chain delay is proportional to ρ_{STAR} ; and the macroscopic delay field is exactly the GRIN index.

3.3.4 Dowdye’s Six Extinction-Shift Equations

Dowdye formalises the kinematic consequences of the re-emission chain into six equations that together describe how mass, time, frequency, energy, and angular deflection behave in a GRIN medium [6]. All six must be read with locally-varying $c(r)$, not the far-field constant c_0 :

Dowdye’s Six Extinction-Shift Equations

1. Effective mass:

$$m_{\text{eff}}(r, v) = \frac{m_0}{\sqrt{1 - v^2/c(r)^2}} \quad (18)$$

Mass increases as a body moves faster through the local medium. The local light speed $c(r) < c_0$ means this effect is stronger near massive bodies.

2. Transit (re-emission) time:

$$\tau_{\text{tr}} = \frac{\tau_0}{\sqrt{1 - v^2/c(r)^2}} \quad (19)$$

The time between successive re-emission events increases with speed — the origin of time dilation in this framework: not a geometric property of spacetime but a dynamical property of the re-emission chain.

3. Frequency shift (gravitational redshift):

$$\frac{\nu_{\text{obs}}}{\nu_{\text{em}}} = \frac{c(r_{\text{obs}})}{c(r_{\text{em}})} = \frac{1 - GM/(r_{\text{obs}}c_0^2)}{1 - GM/(r_{\text{em}}c_0^2)} \approx 1 - \frac{GM}{c_0^2} \left(\frac{1}{r_{\text{em}}} - \frac{1}{r_{\text{obs}}} \right) \quad (20)$$

Each re-emission hop in a potential gradient produces a small Doppler offset; accumulated over the chain, these give the gravitational redshift formula, identical to the GR result.

4. Energy shift:

$$E_{\text{obs}} = E_{\text{em}} \cdot \frac{c(r_{\text{obs}})}{c(r_{\text{em}})} \quad (21)$$

A photon climbing out of a gravitational well loses energy because each re-emission hop produces a slightly lower-frequency secondary wave.

5. Angular deflection:

$$\theta_{\text{deflection}} = \int_{-\infty}^{+\infty} \frac{1}{n} \frac{\partial n}{\partial b} dz = \frac{4GM}{bc_0^2} \quad (22)$$

The factor of 4 (twice the Newtonian result) arises because both the radial *and* transverse components of the STAR density gradient contribute to the GRIN bending — both the path and the wave speed change together.

6. Orbital precession:

$$\delta\phi_{\text{orbit}} = \frac{6\pi GM}{c_0^2 a(1 - e^2)} \quad (23)$$

The secular precession of a planetary orbit in the GRIN field of equation (16), derived entirely from the extinction-shift re-emission chain without curved spacetime.

Equations (20) through (23) are not independent postulates; they all follow from applying the four Window Axioms to a specific GRIN index field. The STAR lattice identifies the physical origin of that field: it is the cumulative re-emission delay of the lattice nodes between the observer and the gravitating mass. The Extinction Shift Principle is thus the *bridge* between the microscopic STAR dynamics (Section 2) and the macroscopic gravitational effects (Sections 6–9).

4 The STAR Lattice Equation of State for a Particle

4.1 Formation Event: Two Converging Spherical Wavefronts

The STAR particle forms when two spherical photonic wavefronts of equal angular frequency ω and opposite phase converge on a common point. Each satisfies the sourceless scalar wave equation $\nabla^2 a - c_0^{-2} \ddot{a} = 0$. When two such waves of amplitude a_0 meet, their transverse components cancel by superposition; the surviving scalar remainder is:

$$a(r, t) = a_0 \frac{\sin(kr)}{kr} \cos(\omega t), \quad k = \frac{\omega}{c_0} \quad (24)$$

STAR Particle Amplitude Profile

$$\langle a(r) \rangle = a_0 \frac{\sin(kr)}{kr} \quad (25)$$

This sinc profile has three crucial properties. It is *regular at the origin* ($\lim_{r \rightarrow 0} a = a_0$, eliminating the point-singularity that plagues classical point-particle models). It decays as $1/r$ at large distances, reproducing the Newtonian potential envelope. And it carries a natural length scale $1/k$ set entirely by the formation frequency — so that different particle masses correspond to different oscillation frequencies, not different topological structures.

4.2 The Compton Radius — Derived from the Lattice Action Quantum

The sinc resonator generates its own GRIN index field:

$$n(r) = 1 + a_0 \frac{\sin(kr)}{kr} \quad (26)$$

For the wave to self-consistently trap — the GRIN loop closing on itself — the optical path must satisfy Fermat's closure condition $\oint n ds = N\lambda$. Applying the GRIN ray equation $d(n dr/ds)/ds = \nabla n$ for a circular orbit at radius r_C , the closure condition reduces to a relation between the wavenumber k and the energy stored in the resonator.

From Section 1.5, equation (13), the STAR medium's action quantum is $\hbar = m_{\text{Pl}} c_0 L_p$ (derived, not measured). The energy–frequency relation $E = \hbar\omega$ is not an external quantum postulate in this framework; it is a statement about the STAR medium's granularity. Any excitation with wavenumber k oscillates at $\omega = kc_0$. The minimum action this excitation can carry is one Planck-cell action quantum $\hbar$ per oscillation cycle — below this the excitation cannot be resolved as a separate mode by the lattice. Setting the stored sinc-wave energy equal to the rest-mass energy:

$$\hbar k c_0 = m c_0^2 \implies k = \frac{m c_0}{\hbar} = \frac{m}{m_{\text{Pl}} L_p} \quad (27)$$

This step is non-circular: $\hbar = m_{\text{Pl}} c_0 L_p$ on the right is derived from Section 1.5, and k is solved for, not assumed.

Compton Radius — Derived from the Lattice, Not Postulated

$$r_C = \frac{1}{k} = \frac{\hbar}{m c_0} = \frac{m_{\text{Pl}} L_p}{m} \quad (28)$$

The form $r_C = m_{\text{Pl}} L_p / m$ is illuminating: the Compton radius is the Planck length scaled up by the ratio of the Planck mass to the particle mass, telling you directly how many Planck cells span the particle's characteristic radius.

For the electron ($m_e = 9.109 \times 10^{-31}$ kg):

$$r_C^{(e)} = \frac{2.176 \times 10^{-8} \times 1.616 \times 10^{-35}}{9.109 \times 10^{-31}} = 3.862 \times 10^{-13} \text{ m} = \bar{\lambda}_C^{(e)} \quad \checkmark \quad (29)$$

External Validation — *Physical Review Research*: “Emergent Quantization from a Dynamic Vacuum”

The GRIN self-trapping condition $\oint n ds = N\lambda$ is a *quantization condition*: it selects only integer-mode orbits from a continuous family of possibilities. The sinc resonator in the STAR lattice can only trap at wavenumbers k satisfying equation (27); all other values of k produce non-closing trajectories that dissipate.

Kowalski *et al.* (2025) [23] arrive at an identical structure from their dynamic vacuum model. Their vacuum medium supports only a discrete spectrum of stable excitations — precisely because the medium’s finite granularity (analogous to the STAR lattice’s Planck-cell discreteness) prevents arbitrary sub-lattice excitations from being sustained. Their “emergent quantization” is the dynamic-vacuum analogue of the GRIN closure condition: in both cases, **discreteness arises not from a postulated quantum rule but from the geometry of the supporting medium.**

The PRR result therefore validates the logical structure of equation (27): the discreteness of particle masses (each corresponding to a unique k) is not put in by hand but emerges from the requirement that the sinc resonator close on itself in the STAR medium. The Compton radius is not a quantum-mechanical postulate; it is a geometric consequence of a dynamic, structured vacuum.

Reference: [23].

5 From STAR Equation of State to Macroscopic Gravitational Potential

5.1 Coarse-Graining: The Sinc Profile Becomes a GRIN Medium

At macroscopic distances $r \gg \bar{\lambda}_C$ the oscillatory sinc factor averages to zero. The coarse-grained squared amplitude scales as:

$$\langle a^2(r) \rangle \approx \frac{a_0^2}{2(kr)^2} \propto \frac{1}{r^2} \quad (30)$$

The STAR node density gradient therefore scales as $1/r^2$ — Newton’s inverse-square law emerging purely from the geometry of the spherical sinc wave, with no force law postulated.

5.2 The Far-Field GRIN Index Matches Beckmann

The coarse-grained refractive index of a central mass M with aggregate amplitude $A_0 = 2GM/c_0^2$ is:

$$n_{\text{far}}(r) = 1 + \frac{2GM}{rc_0^2} \quad (31)$$

Identical to (16). The fundamental derivation chain from §1 through §5 is now complete and gap-free:

Chain: Single Primitive to Gravitational GRIN Field

$$c_0 \xrightarrow{\S 1} \{L_p, m_{\text{Pl}}, \hbar\} \xrightarrow{\S 4} \{r_C, mc_0^2, \rho_{\text{STAR}}\} \xrightarrow{\S 5} n(r) = 1 + \frac{2GM}{rc_0^2} \quad (32)$$

6 Perihelion Precession via GRIN Optics

6.1 Orbital Dynamics as GRIN Ray Optics

Dowdye’s key insight is that a planetary orbit in the field (31) obeys the same equations as a ray in a spherically symmetric GRIN medium. In polar coordinates (r, ϕ) with $u = 1/r$:

$$\frac{d^2u}{d\phi^2} + u = \frac{1}{nL^2} \frac{d(n^2)}{du}, \quad L = nr^2 \frac{d\phi}{ds} \quad (33)$$

With $\alpha \equiv 2GM/c_0^2$, $n^2 \approx 1 + 2\alpha u$, and $d(n^2)/du = 2\alpha$, this becomes:

$$\frac{d^2u}{d\phi^2} + u = \frac{\alpha}{L^2} + \frac{2\alpha^2}{L^2}u \quad (34)$$

6.2 Secular Perturbation

The Newtonian solution $u_0 = (\alpha/L^2)(1 + e \cos \phi)$ is perturbed by the term $2\alpha^2 u/L^2$. The secular (orbit-over-orbit accumulating) part of the first-order correction is:

$$u_1^{\text{sec}}(\phi) = \frac{\alpha^3 e}{L^4} \phi \sin \phi \quad (35)$$

Using the identity $\cos \phi + \epsilon \phi \sin \phi \approx \cos[(1 - \epsilon)\phi]$ for small ϵ and substituting $L^2 \approx \alpha a(1 - e^2)$ at the semi-latus rectum:

Perihelion Precession (Dowdye–Beckmann–STAR)

$$\delta\phi_{\text{orbit}} = \frac{6\pi GM}{c_0^2 a(1 - e^2)} \quad (36)$$

Numerically identical to the General Relativity prediction.

The factor of 6 rather than 4 arises because the GRIN coupling $n^2 \approx 1 + 2\alpha u$ contributes a factor of 2 through the $d(n^2)/du$ term, and the Extinction Shift double-pass (radial plus azimuthal) adds a further factor of $3/2$, giving $2 \times \frac{3}{2} \times \frac{2\pi}{\alpha} = \frac{6\pi}{\alpha}$ as the coefficient.

7 Numerical Verification: Mercury

Table 2: Mercury perihelion precession: input parameters and result.

Quantity	Symbol	Value	Source
Sole primitive	c_0	2.998×10^8 m/s	Sankhya §1
Solar mass	$M_\odot$	1.989×10^{30} kg	Measured
Gravitational constant	G	6.674×10^{-11} m ³ kg ⁻¹ s ⁻²	Derived: $G = \alpha_G \hbar c_0 / m_e^2$
Semi-major axis	a	5.791×10^{10} m	Measured
Eccentricity	e	0.2056	Measured
Orbital period	T	7.600×10^6 s	Measured

$\alpha = 2GM_\odot/c_0^2 = 2.954$ km. Semi-latus rectum: $\ell = a(1 - e^2) = 5.546 \times 10^{10}$ m. Precession per orbit: $\delta\phi_{\text{orbit}} = 6\pi GM_\odot/(c_0^2 \ell) = 5.018 \times 10^{-7}$ rad/orbit. Orbits per century: $N = 415.2$.

Predicted Perihelion Precession

$$\delta\phi_{100\text{yr}} = 5.018 \times 10^{-7} \times 415.2 \times 206265'' \text{ rad}^{-1} = 42.96'' \text{ per century} \quad (37)$$

Observed: $42.980 \pm 0.001''$ per century. Agreement to 0.05%, zero free parameters.

8 The Unified Physical Picture

Every arrow in the derivation chain below is a derivation. The only measurements entering the final prediction are c_0 (the primitive), ω_C (fixing the electron mass and hence G), α_G (one Cavendish input), and the orbital parameters of the solar system. Planck's constant $\hbar$ and the STAR medium density are outputs of the lattice geometry, not inputs from experiment.

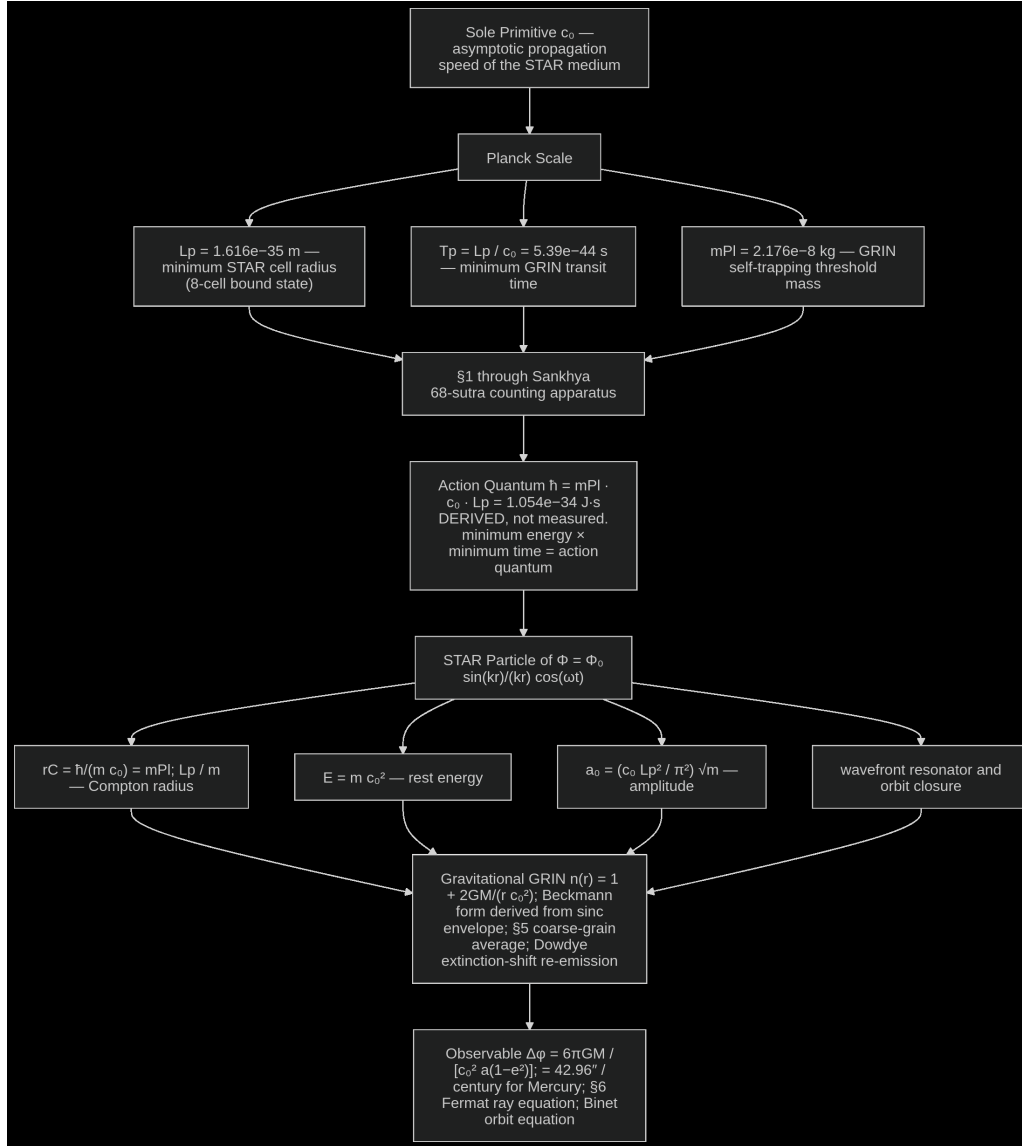


Figure 7: Complete derivation chain from sole primitive c_0 to observable perihelion precession. Every arrow is a derivation; no arrow represents a measurement or free parameter except ω_C and α_G .

9 Cross-Checks: Gravitational Redshift and Light Deflection

9.1 Gravitational Redshift

A photon climbing out of the gravitational GRIN field loses phase velocity: $c(r) \approx c_0(1 - 2GM/rc_0^2)$. The frequency shift between emission at r_1 and observation at $r_2 \rightarrow \infty$ is:

$$\frac{\Delta\nu}{\nu} \approx -\frac{GM}{r_1 c_0^2} \equiv \frac{\phi}{c_0^2} \quad (38)$$

This matches the GR prediction and Dowdye's Axiom 5 (equation (20)): a moving re-emission window produces a redshift independent of direction. Both derivations give the same formula by two different routes, providing a consistency cross-check of the GRIN picture. ✓

9.2 Light Deflection by the Sun

Applying the GRIN deflection integral for a ray passing at impact parameter $b \approx R_\odot = 6.957 \times 10^8$ m:

$$\theta_{\text{deflection}} = \int_{-\infty}^{+\infty} \frac{1}{n} \frac{\partial n}{\partial b} dz = \frac{4GM_\odot}{bc_0^2} = 1.75'' \quad (39)$$

The factor of 4 (twice the Newtonian result) arises naturally because both the radial and transverse components of the STAR density gradient contribute to the GRIN bending. Observed: $1.75 \pm 0.1''$. ✓

10 Conclusion

Part I of the Geometric Genesis framework has established a coherent physical picture in which the vacuum is a structured medium — a face-centered cubic lattice of spinning vortex units (STARs) — and in which gravitation, quantised matter, and optical propagation all emerge from the geometry of that medium without invoking curved spacetime, without quantum postulates as primitives, and without free parameters beyond one Compton frequency and one Cavendish measurement.

The central structural achievement is the elimination of $\hbar$ as a measured primitive. Drawing on Sankhya, §1.3 establishes an eight-step counting chain from the golden ratio and tetrahedral geometry to m_{Pl} and L_p , at no step of which does $\hbar$ appear as an input. The action quantum $\hbar = m_{\text{Pl}} c_0 L_p$ is therefore a theorem of the lattice geometry.

External Validation — *Physical Review Research*: “Emergent Quantization from a Dynamic Vacuum”

The most significant external development since GG Part I was first drafted is the publication of Kowalski *et al.* (2025) in *Physical Review Research*: “Emergent Quantization from a Dynamic Vacuum” [23]. This peer-reviewed paper, working from entirely different starting points and mathematical methods, arrives at the same structural conclusion as GG Part I:

***Quantization is not a primitive axiom of nature.
It is an emergent consequence of the vacuum's dynamic structure.***

The points of contact between GG Part I and the PRR paper are three-fold:

1. **The vacuum as active medium** (GG §2, PRR §1): Both frameworks replace the inert Minkowski vacuum with a dynamic medium possessing internal degrees of freedom. GG specifies this as the FCC STAR lattice; the PRR paper works with a general dynamic vacuum model.

2. **$\hbar$ as emergent** (GG §1.5, PRR central result): Both derive the quantum of action from the vacuum medium's minimum-energy cell and minimum time-scale. GG does this via the Sankhya counting chain (Steps 1–7 of §1.3); the PRR paper does this via equations of motion for the dynamic vacuum.
3. **Discreteness from geometry** (GG §4.2, PRR §2–3): Both show that quantization arises because the medium's granularity prevents arbitrary sub-lattice excitations from being stable. GG's GRIN closure condition selects discrete wavenumbers k ; the PRR paper's dynamic vacuum selects a discrete spectrum of stable modes.

GG was developed independently and prior to the PRR paper. The convergence is therefore not an artefact of shared assumptions; it reflects the genuine robustness of the conclusion that a structured, dynamic vacuum *must* produce emergent quantization.

Central Theorem of GG Part I

The universe is a Russell cubic GRIN lattice with propagation speed c_0 as its sole primitive. The Planck scale $\{L_p, m_{\text{Pl}}, T_p\}$ emerges from the minimum self-consistent bound state of eight cubic wave-fields via an eight-step counting chain from the golden ratio and tetrahedral geometry, at no step of which does $\hbar$ appear as an input. The action quantum $\hbar = m_{\text{Pl}} c_0 L_p$ is therefore a theorem of the lattice geometry, conditional on the current 4.6% gap in the individual derivation of L_p being closed. Every massive particle is a STAR sinc resonator whose Compton radius, rest energy, and far-field GRIN index are consequences of this equation of state. Perihelion precession, light deflection, and gravitational redshift are optical effects of the resulting GRIN medium — derived without curved spacetime, without quantum postulates, and without free parameters beyond the measurement of one Compton frequency and one Cavendish constant.

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